

SQL ASSIGNMENT – COVID DATA ANALYSIS

Dataset:

You are given an Excel file named Global_Covid_Data.xlsx with a sheet CovidData. Columns include:

Province, Country, Latitude, Longitude, Date, Confirmed, Deaths, Recovered, Active, Incidence_Rate, Case_Fatality_Ratio.

PART 0 — DATA IMPORT & CLEANING

1. Import the Excel file into SQL Server as CovidData.
2. Check all rows containing NULL values.
3. Replace NULLs in all numeric columns with 0.

PART 1 — BASIC EXPLORATION

1. Display first 10 rows.
2. Count total number of rows.
3. Count distinct months per year.
4. Show start and end date.
5. Count rows per year-month.

PART 2 — DESCRIPTIVE STATISTICS

Compute monthly min, max, and total of Confirmed, Deaths, Recovered.

PART 3 — CENTRAL TENDENCY

Compute monthly mean.

Compute median of Confirmed.

Compute mode of Confirmed per month.

PART 4 — DISPERSION

Compute total, average, variance, std deviation for Confirmed, Deaths, Recovered (overall & by month-year).

PART 5 — PERCENTILES

Compute 50th, 60th, 90th, 95th percentiles using PERCENTILE_DISC or PERCENTILE_CONT.

PART 6 — TOP RECORDS

Top 10 highest Confirmed, Deaths, Recovered.

PART 7 — CORRELATION

Correlation between:

- Confirmed vs Deaths
- Confirmed vs Recovered
- Deaths vs Recovered

PART 8 — RANKING

Use ROW_NUMBER() to rank by Confirmed cases.

PART 9 — REGRESSION MODEL

Compute slope and intercept where:

x = Confirmed

y = Deaths

Write regression equation $y = mx + b$ and interpret it.

Deliverables:

1. SQL script (.sql)
2. Final report (Word/PDF) with results and insights.